KG COLLEGE OF ARTS AND SCIENCE
Autonomous Institution | Affiliated to Bharathiar University
Accredited with A++ Grade by NAAC
ISO 9001:2015 Certified Institution
KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Hospital Bed Allocation and Availability Monitoring System

Submitted by

VETRISELVAN A

25BIT061

Under the Guidance of

DHILEEBAN A

Technical Trainer

Technical Department, KG College of Arts and Science

Academic Year: **2025 – 2026**

APRIL 2026 Semester: **I**

DECLARATION

I hereby declare that the project entitled " **"Hospital Bed Allocation and Availability Monitoring System"** submitted in partial fulfillment of the requirements for the course **JAVA PROGRAMMING** is a **record of my original work** carried out under the guidance of , Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

VETRISELVAN A

(Your Name)

CERTIFICATE

This is to certify that the project entitled **" "Hospital Bed Allocation and Availability Monitoring System"** is a **bona fide work** carried out by **[VETRISELVAN A] [I-BSc IT-.'A']**, [**]**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **[JAVA PROGRAMMING]**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: COIMBATORE

Date: 09.04.2026

Submitted for the Viva-Voce Examination held on **KG COLLAGE OF ARTS & SCIENCE**

Guide:
(Technical Trainer, KGCAS)

Head of the Department
(Technical Head, KGCAS)

TABLE OF CONTENTS

S. No.	Topics	Page. No.
1	**Introduction**	
1.1	Objective	
1.2	Overview	
1.3	Features	
2	**Program Analysis**	
2.1	Problem Definition	
2.2	Project Overview	
2.3	Scopes and Limitations	
3	**System Design**	
3.1	Modules	
4	**Methodology**	
4.1	Problem Analysis	
4.2	System Design	
4.3	Program Development	
4.4	Testing and Validation	
4.5	Result and Documentation	
4.6	Algorithm	
4.7	Flowchart	
5	**Source Code**	
6	**Input and Output Screens**	
7	**Discussion and Conclusion**	
7.1	Discussion	
7.2	Conclusion	
8	**References**	

ABSTRACT

The Hospital Bed Allocation and Availability Monitoring System is a Java-based application designed to efficiently manage and monitor hospital bed resources. In many healthcare facilities, bed allocation is handled manually, which can lead to delays, errors, and inefficient utilization of available resources. This system aims to overcome these challenges by providing an automated and organized approach to managing bed availability.

The application maintains records of beds, including their type and occupancy status, as well as patient details. It allows hospital staff to allocate beds to patients based on availability and ward type, release beds when patients are discharged, and view the current status of all beds in real-time. This ensures quick decision-making and improves patient care services.

Developed using Core Java concepts such as classes, objects, collections, loops, and conditional statements, the system provides a simple and effective solution for small and medium-scale hospitals. The current implementation is console-based, but it can be further enhanced into a GUI-based or web-based application with database integration.

The main objective of this system is to improve efficiency in hospital management, reduce manual errors, and ensure optimal utilization of bed resources

1. INTRODUCTION

In today's digital world, managing information efficiently is very important. People create notes for studies, work, and personal tasks. However, traditional note-taking methods such as paper notebooks or scattered files make it difficult to organize and retrieve information.

The **Hospital Bed Allocation and Availability Monitoring System** is a Java-based system that provides a structured way to manage notes digitally. It allows users to create, store, edit, and delete notes, as well as attach files to them.

This system simplifies note management and ensures that all information is stored securely and can be accessed easily whenever needed.

1.1 Objective

The main objective of the Hospital Bed Allocation and Availability Monitoring System is to efficiently manage hospital bed resources. It helps in tracking available and occupied beds in real-time, reducing delays in patient admission and improving hospital service management.

1.2 Overview

This system is a Java-based application designed to automate the process of bed allocation in hospitals. It maintains records of patient details, bed availability, and allocation status. The system ensures that hospital staff can quickly assign beds to patients without manual errors.

1.3 Features

Real-time bed availability tracking

Patient registration and management

Automatic bed allocation

Bed status update (occupied/vacant)

Easy data retrieval and display

Simple and user-friendly interface

2. PROGRAM ANALYSIS

2.1 Problem Definition

In many hospitals, bed allocation is done manually, which leads to confusion, delays, and inefficient utilization of resources. There is a need for a system that can automate and simplify this process.

2.2 Project Overview

This project helps hospital staff manage bed allocation digitally. It stores patient information and bed details, making it easy to assign beds and monitor their availability efficiently.

2.3 Scopes and Limitations

Scope

Can be used in small and medium hospitals

Helps reduce manual errors

Improves patient service

Limitations

Console-based system (no GUI)

Limited to local data storage

Does not support online access

3. SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The **Digital Note Organizer with File Storage** is designed as a menu-driven application where users can select food items, enter quantities, and generate a final bill automatically.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system.

System design defines the structure and working of the system.

The **Digital Note Organizer with File Storage** is divided into modules for better organization.

3.3.1 Modules

Patient Module – Stores patient details

Bed Management Module – Tracks bed availability

Allocation Module – Assigns beds to patients

Display Module – Shows current status

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the **Digital Note Organizer with File Storage** , a structured development approach was followed to ensure that the system meets the requirements and functions efficiently.

The development process includes the following stages:

• Problem Analysis
• System Design
• Program Development
• Testing and Validation
• Result and Documentation

The DDRMS follows a structured development approach.

4.1 Problem Analysis

The system identifies inefficiencies in manual bed allocation and provides a structured digital solution.

4.2 System Design

The system is designed using object-oriented programming concepts like classes and objects in Java.

4.3 Program Development

The program is developed using Java with basic concepts like loops, arrays, and conditional statements.

4.4 Testing and Validation

The system is tested with different inputs to ensure correct allocation and proper display of results.

4.5 Result and Documentation

The system successfully allocates beds and displays availability. All outputs are documented properly.

4.6 Algorithm

Start

Initialize total number of beds

Display menu options

Get user choice

If choice = Add Patient

Enter patient details

Assign available bed

If choice = View Beds

Display available and occupied beds

If choice = Exit

Stop program

End

4.7 Flowchart (Text Representation)

Start → Display Menu →

→ Enter Choice →

→ Add Patient / View Beds →

→ Process Data →

→ Display Output →

→ Exit

5. SOURCE CODE

```
class Bed{
    int bedId;
    String wardType;
    boolean isOccupied;

    public Bed(int bedId, String wardType) {
        this.bedId = bedId;
        this.wardType = wardType;
        this.isOccupied = false;
    }
}


class Patient {
    int patientId;
    String name;
    int allocatedBedId;
```

```
    public Patient(int patientId, String name) {
        this.patientId = patientId;
        this.name = name;
    }
}


import java.util.*;

class Hospital {
    List<Bed> beds = new ArrayList<>();
    List<Patient> patients = new ArrayList<>();

    public void addBed(int id, String type) {
        beds.add(new Bed(id, type));
    }

    public void addPatient(int id, String name) {
        patients.add(new Patient(id, name));
    }

    public void allocateBed(int patientId, String wardType) {
        for (Bed bed : beds) {
            if (!bed.isOccupied && bed.wardType.equalsIgnoreCase(wardType)) {
                for (Patient p : patients) {
                    if (p.patientId == patientId) {
                        bed.isOccupied = true;
                        p.allocatedBedId = bed.bedId;
```

```
                System.out.println("Bed " + bed.bedId + " allocated to " + p.name);
                return;
            }
        }
    }
    System.out.println("No available bed in " + wardType);
}

public void releaseBed(int patientId) {
    for (Patient p : patients) {
        if (p.patientId == patientId && p.allocatedBedId != -1) {
            for (Bed b : beds) {
                if (b.bedId == p.allocatedBedId) {
                    b.isOccupied = false;
                    System.out.println("Bed " + b.bedId + " released from " + p.name);
                    p.allocatedBedId = -1;
                    return;
                }
            }
        }
    }
    System.out.println("Patient not found or no bed allocated");
}

public void showAvailableBeds() {
    System.out.println("Available Beds:");
    for (Bed b : beds) {
        if (!b.isOccupied) {
```

```
                System.out.println("Bed ID: " + b.bedId + " | Type: " + b.wardType);
            }
        }
    }
}

public class HospitalSystem {
    public static void main(String[] args) {
        Hospital hospital = new Hospital();

        hospital.addBed(1, "ICU");
        hospital.addBed(2, "General");
        hospital.addBed(3, "Private");
        hospital.addBed(4, "ICU");

        hospital.addPatient(101, "Vetri");
        hospital.addPatient(102, "Alvin");

        hospital.allocateBed(101, "ICU");
        hospital.allocateBed(102, "General");

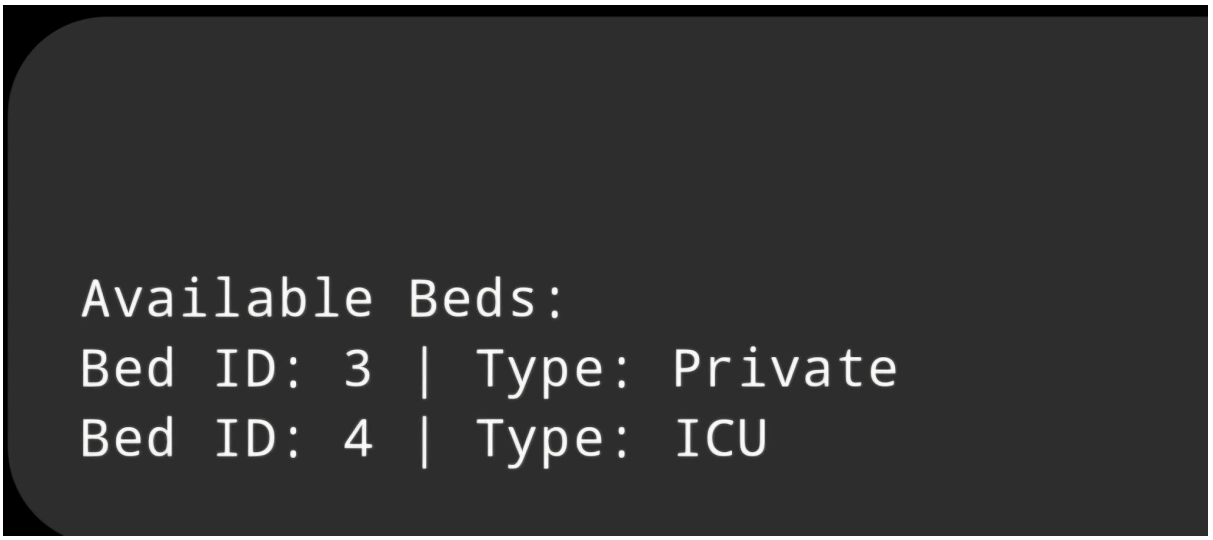
        hospital.showAvailableBeds();

        hospital.releaseBed(101);

        hospital.showAvailableBeds();
    }
}
```

6. INPUT AND OUTPUT SCREENS






Bed 1 released from Vetri

```
Available Beds:
Bed ID: 1 | Type: ICU
Bed ID: 3 | Type: Private
Bed ID: 4 | Type: ICU
```

```
Bed 1 allocated to Vetri
Bed 2 allocated to Alvin
Available Beds:
Bed ID: 3 | Type: Private
Bed ID: 4 | Type: ICU
Bed 1 released from Vetri
Available Beds:
Bed ID: 1 | Type: ICU
Bed ID: 3 | Type: Private
Bed ID: 4 | Type: ICU
```

7. DISCUSSION & CONCLUSION

7.1 Discussion

The system simplifies the process of managing hospital beds. It reduces manual effort and ensures accurate allocation. The use of Java makes it simple and efficient for small-scale applications.

7.2 Conclusion

The Hospital Bed Allocation System is an effective solution for managing hospital resources. It improves efficiency, reduces errors, and enhances patient care. The system can be further improved by adding a graphical interface and database connectivity.

8. REFERENCES

The following resources were used during the development of this project:

Books

• Herbert Schildt – *Java: The Complete Reference*

Websites

• https://www.oracle.com/java
• https://www.geeksforgeeks.org
• https://www.w3schools.com
• https://www.javatpoint.com